Multiple Choice Questions (MCQs)

1. What does a cell in a database table contain?

a) A complete record about one entity

b) A collection of related data

c) One value at the intersection of a row and a column

d) Multiple records

2. The statement "Patient #1001 has a body temperature of 38.5°C" is an example of:

a) Data

b) Information

c) Knowledge

d) A database

3. Applying human understanding, experience, and rules to information to make a decision is called:

a) Data

b) Information

c) Knowledge

d) Persistence

4. Which of the following is an example of a Relational Database?

a) MongoDB

b) Firebase

c) PostgreSQL

d) CSV

5. Which tier in the 3-tier architecture is responsible for User Experience & Rendering (formatting HTML, CSS, images)?

a) Tier 1 (Client Side)

b) Tier 2 (Application/Logic)

c) Tier 3 (Data)

d) None of the above

6. Which tier writes and reads data from relational or NoSQL databases?

a) Presentation Tier

b) Logic Tier

c) Data Tier

d) Network Tier

7. What is the primary purpose of the INT or INTEGER data type?

a) To store text

b) To store standard whole numbers

c) To store dates

d) To store exact decimals

8. Which data type should be used for exact decimals where precision is critical, such as money?

a) FLOAT

b) DOUBLE

c) DECIMAL

d) REAL

True/False (T/F)

20. A table is a way to organize data inside a database and consists of rows and columns.

21. In a database table, a row represents one property, such as a student's name.

22. "Data" is defined as disconnected values without context (e.g., 38.5, 1001, True).

23. Flat files and spreadsheets (like CSV and Excel) slow down severely when dealing with millions of rows.

24. A Primary Key allows duplicate values so you can group similar records together.

25. A Primary Key does not allow NULL values.

26. The client side in a 3-tier architecture is trusted for security validations.

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41. If you need to store student names in Arabic or other languages in SQL Server, which suggested data type should you choose?

a) INT

b) VARCHAR

c) NVARCHAR

d) DATE

42. "A store cashier can read price tags and record sales, but only the HR manager can view employee salary records." This is an example of which database advantage?

a) Persistence

b) Concurrency

c) Security (Access Control)

d) Data Integrity

43. What happens in a database system if two people try to book seat 12A on a flight at the exact same time?

a) Both users are given the seat.

b) The database engine locks the row instantaneously, guaranteeing only one booking succeeds.

c) The database crashes and restarts.

d) The record is duplicated.

44. Which numeric data type is best used for scientific measurements where extreme precision isn't required, but performance is?

a) DECIMAL

b) NUMERIC

c) INT

d) FLOAT / REAL

45. If a Student table has a Subjects column containing the value "Math, English, Python" for a single record, which normal form does this violate?

a) 1NF

b) 2NF

c) 3NF

d) None

True/False (T/F)

59. According to the text, the Client Side is responsible for calculating final discounts and determining user authorization levels.

60. Transactional Integrity in Tier 3 ensures ACID compliance (e.g., ensuring money is deducted from Account A and added to Account B together, or neither).

61. UUIDs / GUIDs are universally unique identifiers containing 36-character strings used for unique primary keys across distributed systems.

62. A table can satisfy 2NF and still contain unnecessary data duplication if it has transitive dependencies.

63. Under 3NF, every non-key attribute should depend on the key, the whole key, and nothing but the key.

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71. Suppose a table has a composite Primary Key (A, B) and non-key attributes C and D.

If attribute C depends on the full key (A, B), but attribute D depends only on B, which Normal Form does this table violate?

a) 1NF

b) 2NF

c) 3NF

d) 4NF

72. You have a table with EmployeeID (Primary Key), EmployeeName, JobID, and JobTitle.

If JobTitle depends on JobID, and JobID depends on EmployeeID, which Normal Form is violated?

a) 1NF

b) 2NF

c) 3NF

d) None

73. According to the text, which process in Tier 2 involves calling external APIs (e.g., Stripe for payments, Twilio for SMS)?

a) Input Validation

b) Response Construction

c) Integration & Orchestration

d) Business Rules Enforcement

74. A StudentSubjects table has a composite primary key (StudentID, SubjectID) and a non-key attribute Grade. If Grade depends on both StudentID and SubjectID together, what is true about the table regarding 2NF?

a) It violates 2NF.

b) It satisfies 2NF (assuming it is in 1NF) because there is no partial dependency.

c) It violates 3NF.

d) It has a transitive

dependency.

75. If Ahmed (StudentID = 101) enrolls in two subjects, and we store both records in a single unnormalized table, why can't StudentID alone be the primary key?

a) Because StudentID will not be unique for each enrollment.

b) Because Ahmed is not a unique name.

c) Because StudentID allows NULL values.

d) Because primary keys cannot be numbers.

76. Which database concept ensures that a column set to Age will strictly reject negative numbers?

a) Data Persistence

b) Transactional Integrity

c) Data Integrity (Enforcing Correctness)

d) Concurrency

77. You have a table in 2NF with fields: StudentID (PK), StudentName, DepartmentID, and DepartmentName. To convert this to 3NF, how should you structure the tables?

a) Keep it as one table; it is already in 3NF.

b) Create two tables: one with StudentID (PK), StudentName, DepartmentID (FK) and another with DepartmentID (PK) and DepartmentName.

c) Create two tables: one with StudentID and StudentName, and another with StudentName and DepartmentName.

d) Create three tables to separate the name and the department.

78. When breaking down a table with a partial dependency (violating 2NF) into three normalized tables, what happens to the non-key attributes?

a) They are all deleted.

b) They remain in the original table with the partial key.

c) Each non-key attribute now depends on the whole key of its respective new table.

d) They become composite primary keys.

True/False (T/F)

87. If a table has a single-column primary key and currently satisfies 1NF, it is impossible for it to violate 2NF because partial dependencies require a composite key.

88. In database normalization, moving from 2NF to 3NF involves separating information into new tables to eliminate partial dependencies.

89. Because the client side performs lightweight input formatting, the server side does not need to re-validate the incoming data.

90. In a 3-tier architecture, the Data Tier (Tier 3) handles the execution of critical business logic like calculating final discounts.

91. "Data Integrity" guarantees that money is deducted from Account A and added to Account B together, or neither.

92. In the 3NF conversion example provided in the text, the DepartmentID in the Students table acts as a foreign key reference to the Departments table.

93. If you have an unnormalized table where one student studies multiple subjects, creating a composite primary key out of (StudentID, SubjectID) immediately puts the entire table into 3NF.